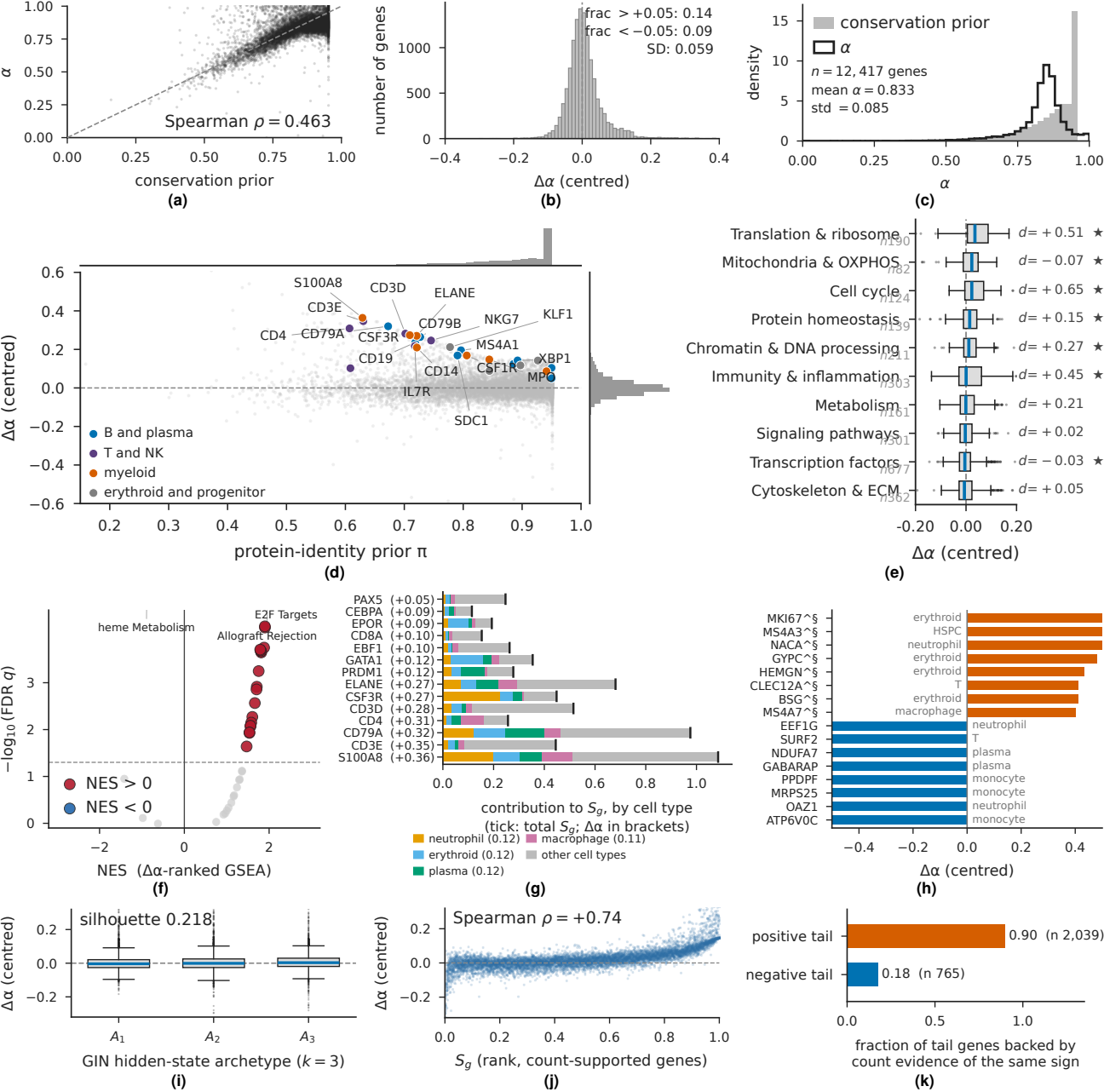

Spearman ρ = 0.463
conservation prior
α
(a)
frac > +0.05: 0.14
frac < −0.05: 0.09
SD: 0.059
number of genes
Δα (centred)
(b)
conservation prior
α
n = 12,417 genes
mean α = 0.833
std = 0.085
density
(c)
S100A8
CD3D
ELANE
CD3E
CD4
CD79A
CSF3R
CD79B
NKG7
KLF1
MS4A1
CD19
CD14
CSF1R
XBP1
MPO
IL7R
SDC1
B and plasma
T and NK
myeloid
erythroid and progenitor
Δα (centred)
protein-identity prior π
(d)
Translation & ribosome n190 d = +0.51 ★
Mitochondria & OXPHOS n82 d = −0.07 ★
Cell cycle n124 d = +0.65 ★
Protein homeostasis n139 d = +0.15 ★
Chromatin & DNA processing n211 d = +0.27 ★
Immunity & inflammation n303 d = +0.45 ★
Metabolism n161 d = +0.21
Signaling pathways n301 d = +0.02
Transcription factors n877 d = −0.03 ★
Cytoskeleton & ECM n362 d = +0.05
Δα (centred)
(e)
heme Metabolism
E2F Targets
Allograft Rejection
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
PAX5 (+0.05)
CEBPA (+0.09)
EPOR (+0.09)
CD8A (+0.10)
EBF1 (+0.10)
GATA1 (+0.12)
PRDM1 (+0.12)
ELANE (+0.27)
CSF3R (+0.27)
CD3D (+0.28)
CD4 (+0.31)
CD79A (+0.32)
CD3E (+0.35)
S100A8 (+0.36)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
neutrophil (0.12)
macrophage (0.11)
erythroid (0.12)
other cell types
plasma (0.12)
(g)
MKI67^§ erythroid
MS4A3^§ HSPC
NACA^§ neutrophil
GYPC^§ erythroid
HEMGN^§ erythroid
CLEC12A^§ T
BSG^§ erythroid
MS4A7^§ macrophage
EEF1G neutrophil
SURF2 T
NDUFA7 plasma
GABARAP plasma
PPDPF monocyte
MRPS25 monocyte
OAZ1 neutrophil
ATP6V0C monocyte
Δα (centred)
(h)
silhouette 0.218
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.74
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail
0.90 (n 2,039)
negative tail
0.18 (n 765)
fraction of tail genes backed by
count evidence of the same sign
(k)